

VG Vibrational Geometrodynamics - SERIES 4, EPISODE 4 :

Spin Dynamics in the Primary Network – The Foundation for Nodal Cavity Frequencies and Inter-Stratum Coupling

Authors: Physicist Pop Rodica & DeepSeek R1 Assistant

Date : 28.07.2026

ABSTRACT

This episode establishes the dynamical backbone of the VG framework by deriving the time evolution of nodal spins from the complete unified Hamiltonian. We show that spin is not a static label but a dynamical field whose evolution generates torsion, modulates the resonant frequency of nodal cavities, and directly affects the inter-stratum coupling tensor. The resulting generalised Landau–Lifshitz equation provides the missing foundation for the Hubble-from-cavity derivation, the reformulation of the Grand Equation, and the topological-gate application. All results are expressed in a formalism fully consistent with VG Series 1–3 and RTS1–25.

Keywords: spin dynamics, Landau–Lifshitz equation, nodal cavity, inter-stratum tensor, torsion field, C_{self} , Primary Network, VG Series 4.

1. INTRODUCTION – Why Spin Must Evolve

In previous VG episodes, spin was treated as a fixed attribute of each node – a compass needle that never turns. While this simplification was sufficient for static geometries, it fails to describe the real network, where couplings relax, coherence changes, and the universe expands and contracts.

In VG Series 3, Episode 7, we established that spin is the root of torsion and gravity. In VG Series 3, Episode 8, we defined coherence as spin alignment. In VG12, we showed that spin acts as a permissivity filter for quantum statistics. In all these cases, spin is the driver – but its own dynamics were never written.

This episode fills that gap. We derive the evolution equation for $\hat{S}_i(t)$ directly from the full Hamiltonian, and we show that once spin moves, everything else follows: cavity frequencies, inter-stratum coupling, torsion, and even the Hubble parameter.

2. THE COMPLETE UNIFIED HAMILTONIAN

We start from the Hamiltonian that combines:

- the isolated nodal oscillator (VG Series 1),
- the elastic and kinetic couplings (VG Series 1),
- the spin-permissivity filter (VG12),
- the spin-orbit torsion term (VG Series 3, Episode 7).

$$H = \sum_i \left(\frac{p_i^2}{2} + \frac{1}{2} \omega_i^2 q_i^2 \right) + \sum_{\langle i,j \rangle} J_{ij} (1 - \hat{S}_i \cdot \hat{S}_j) (q_i - q_j)^2 + \sum_{\langle i,j \rangle} \lambda_{ij} p_i p_j + \sum_{\langle i,j \rangle} \gamma_{ij} (\hat{S}_i \times \hat{S}_j) \cdot q_i p_j$$

Notation:

- $\hat{S}_i$ – spin operator of node i,
- q_i, p_i – dimensionless coordinate and momentum of the nodal oscillator,
- ω_i – bare frequency of node i,
- J_{ij} – elastic coupling,
- λ_{ij} – kinetic coupling,
- γ_{ij} – spin-orbit coupling strength.

3. DERIVATION OF SPIN EVOLUTION

We treat $\hat{S}_i^a$ as Heisenberg operators. The time derivative is:

$$\frac{d\hat{S}_i^a}{dt} = \frac{i}{\hbar} [H, \hat{S}_i^a]$$

After calculating the commutators, we obtain:

$$\frac{d\hat{S}_i}{dt} = \vec{\Omega}_i \times \hat{S}_i + \text{feedback terms}$$

where the effective field $\vec{\Omega}_i$ is:

$$\vec{\Omega}_i = \frac{2}{\hbar} \sum_{j \in \mathcal{N}(i)} J_{ij} (q_i - q_j)^2 \hat{S}_j + \frac{1}{\hbar} \sum_{j \in \mathcal{N}(i)} \gamma_{ij} q_i p_j \hat{S}_j$$

In the simplest approximation, we keep only the precession term:

$$\frac{d\hat{S}_i}{dt} = \vec{\Omega}_i \times \hat{S}_i$$

This is a generalised Landau–Lifshitz equation for the nodal spins.

Note: While the linear precession term is sufficient for deriving the time-dependence of ω_i , the non-linear feedback terms are essential for ensuring finite relaxation times and preventing unphysical divergence of the coupling at the Planck scale. These terms will be crucial for Deep's exact derivation of the Hubble relaxation rate. Additionally, for macroscopic applications (such as Deep's derivation of $H(t)$), the quantum operators $\hat{S}_i$ can be replaced by their expectation values $\langle \hat{S}_i \rangle$ allowing a classical treatment of the network's collective dynamics.

4. CONSEQUENCES FOR THE NETWORK DYNAMICS

4.1. Cavity Frequency Becomes Time-Dependent

$$\omega_i(t) = \frac{1}{\hbar} \left| \sum_{j \in \partial i} J_{ij}^{\text{eff}}(t) \hat{S}_{ij}(t) \right|$$

With spin evolution, $\hat{S}_{ij}(t)$ follows a deterministic trajectory, hence:

$$\omega_i(t) = \omega_i(\hat{S}_i(t), \hat{S}_j(t), q_i(t), p_i(t))$$

4.2. Inter-Stratum Tensor Becomes a Functional of Spin Dynamics

$$\mathcal{T}\alpha\beta(t) = \sum_{i \in C_\alpha, j \in C_\beta} e^{-i\omega_i t} J_{ij}(t) e^{-i\omega_j t}$$

Since $\omega_i(t)$ and $J_{ij}(t)$ are functions of evolving spins:

$$\mathcal{T}\alpha\beta(t) = \mathcal{T}\alpha\beta[\hat{S}_i(t), \hat{S}_j(t)]$$

4.3. Torsion Field as a Spin-Derived Quantity

$$\text{Torsion} \propto \frac{d}{dt} (\hat{S}_i \times \hat{S}_j)$$

5. CONNECTION TO THE OTHER PROJECTS

- **Hubble from Cavity Frequency** : Provides the dynamical equation for $\omega_i(t)$.
- **Grand Equation + Cavity**: Supplies the spin-evolution term for $H_{\text{total}} = H + \mathcal{L}_{vivo}$.
- **Topological Gates** : Gives a time-dependent $\mathcal{T}_{\alpha\beta}$.

6. CONCLUSIONS

We have derived the first dynamical equation for nodal spins in the VG framework. The result is a generalised Landau–Lifshitz equation that couples spin precession to the vibrational state of neighbouring nodes. This equation:

- makes the cavity frequency $\omega_i(t)$ a time-dependent quantity,
- makes the inter-stratum tensor $\mathcal{T}_{\alpha\beta}(t)$ a functional of spin evolution,
- identifies torsion as the rate of change of spin misalignment,
- provides the missing foundation for the three main projects of the team.

Spin is no longer a static label – it is the living engine of the Primary Network.

MATHEMATICAL APPENDIX – Key Steps

A1. Commutator for the spin-permissivity term:

$$[H_{\text{perm}}, \hat{S}^a_i] = i\hbar \sum_j J_{ij} (q_i - q_j)^2 \epsilon^{abc} \hat{S}^b_i \hat{S}^c_j$$

A2. Commutator for the spin-orbit term:

$$[H_{\text{so}}, \hat{S}^a_i] = \sum_j \gamma_{ij} q_i p_j [(\hat{S}_i \times \hat{S}_j)^a, \hat{S}^a_i]$$

A3. Final vector form:

$$\frac{d\hat{S}_i}{dt} = \vec{\Omega}_i \times \hat{S}_i$$